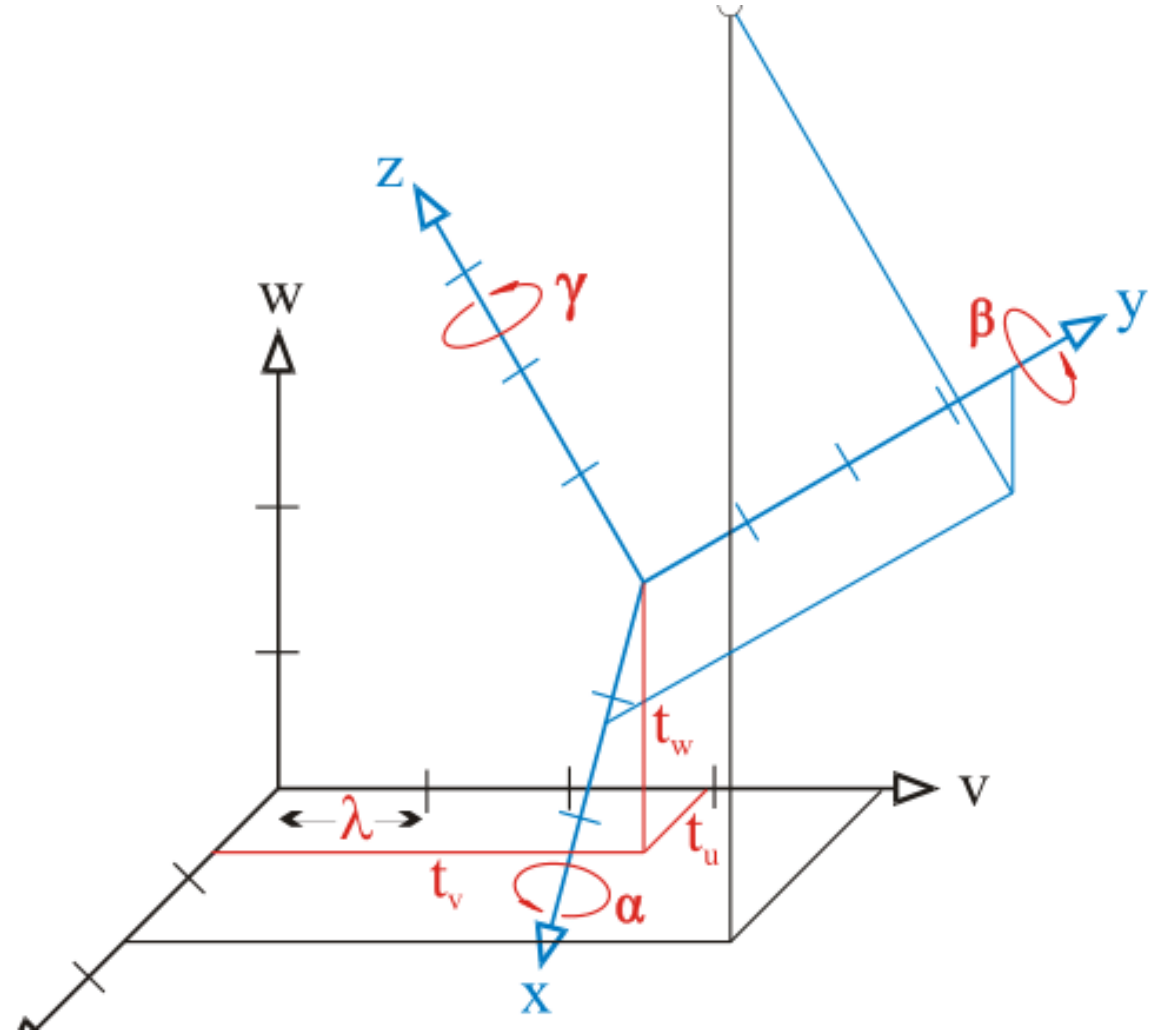
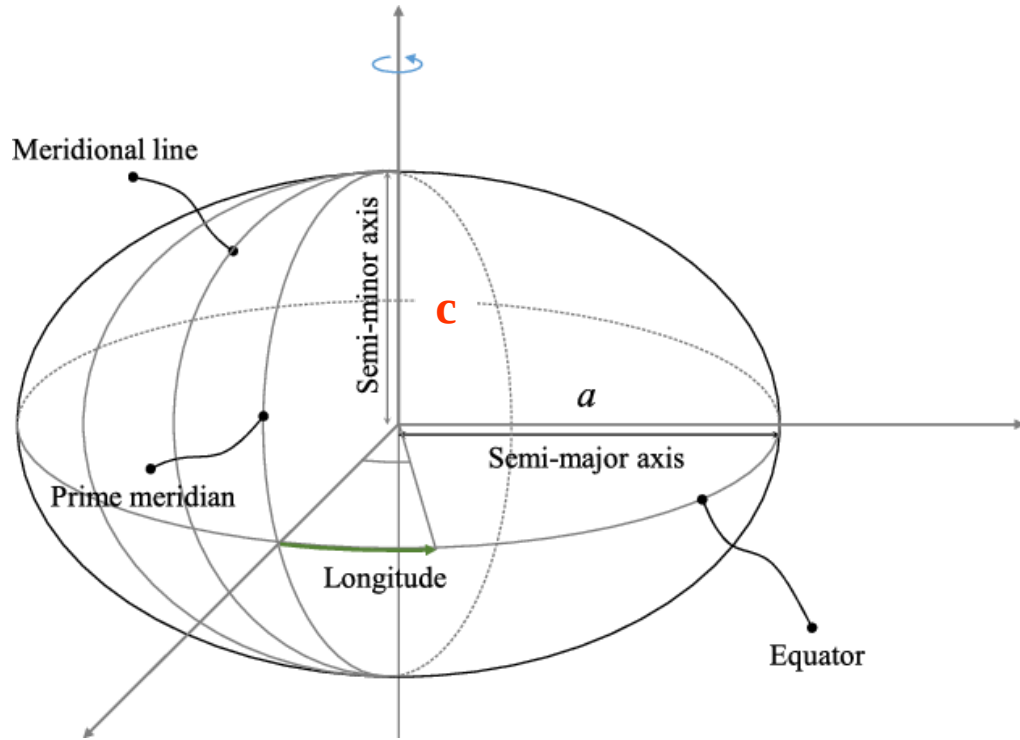


LAB 3: Geodesy and reference system



Prof. MARCO PIRAS

Ellipsoid's parameters



$$\alpha = \frac{a - c}{a}$$

$$e^2 = \frac{a^2 - c^2}{a^2}$$

$$e'^2 = \frac{e^2}{1 - e^2}$$

$$c^2 = a^2(1 - e^2)$$

$$e'^2 = \frac{a^2 - c^2}{c^2}$$

BESSEL (1841)

CLARKE (1880)

HAYFORD (1909)

WGS84 (1984)

$a = 6.377.397 \text{ m}$

$a = 6.378.243 \text{ m}$

$a = 6.378.388 \text{ m}$

$a = 6.378.137 \text{ m}$

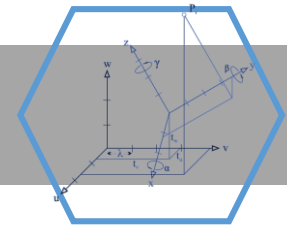
$\alpha = 1/299,2$

$\alpha = 1/293,5$

$\alpha = 1/297,0$

$\alpha = 1/298,257223563$



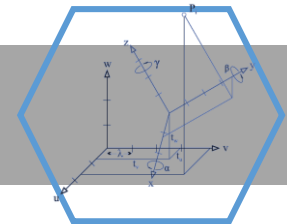


Ex 1: Ellipsoid's parameters

ELLISSOIDE	a [m]	α	$c = a(1 - \alpha)$	e^2	e'^2
Delambre	6.376.985	1/308.6			
Everest	6.377.276	1/300.8			
Bessel	6.377.397	1/299.1528128			
Fisher	6.378.338	1/288.5			
Clarke	6.378.249	1/293.5			
Helmert	6.378.140	1/298.3			
Hayford	6.378.388	1/297.0	6.356.911,9	0,0067226700	0,0067681702
Krassovsky	6.378.245	1/298.3			
WGS84	6.378.137	1/298.257223563			

GOAL: To estimate the ellipsoid's parameters and make a comment of the results, making a comparison.





Ex.2 coordinates transformation from (φ, λ, h) to (XYZ)

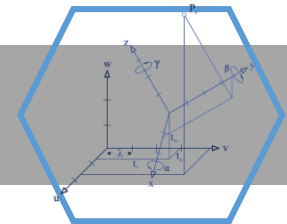
Parametric equation of the ellipsoid

$$\begin{cases} X = \left(\frac{a}{W} + h \right) \cos \varphi \cdot \cos \lambda \\ Y = \left(\frac{a}{W} + h \right) \cos \varphi \cdot \sin \lambda \\ Z = \left[\frac{a}{W} (1 - e^2) + h \right] \cdot \sin \varphi \end{cases}$$

where

$$W = \sqrt{1 - e^2 \cdot \sin^2 \varphi}$$





Ex.2 coordinates transformation from (φ, λ, h) to (XYZ)

points	Latitude (φ)	Longitude (λ)	h (m)
1	44° 45' 01",03930	7° 24' 29",20335	322,490*
2	44° 47' 10",90505	7° 30' 26",53939	305,736

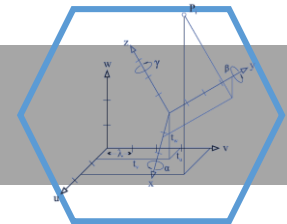
Considering the following ellipsoids:

- WGS 84
- Hayford

GOAL:

- 1) to estimate the coordinates using both ellipsoids
- 2) Try to change «h1», adding 2000 m. How is this value distributed in X, Y, Z?





Ex.3 coordinates transformation (X, Y, Z) to (φ, λ, h)

	X	Y	Z
	[m]	[m]	[m]
1	4498329,37	562840,77	4472537,61
2	4495694,27	592457,86	4470744,78
3	4503484,72	578160,75	4465024,30
4	4499525,43	585034,13	4467910,36

Considering the following ellipsoids:

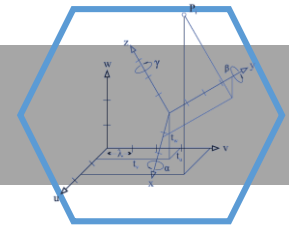
- WGS 84
- Hayford

GOAL: to estimate geographical coordinates

: to define the number of iterations

: to describe the variation of «h» during the procedure



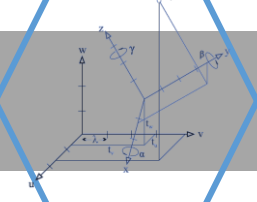


Ex.3 coordinates transformation (X, Y, Z) to (φ, λ, h)

Longitude estimation

$$\lambda = \arctg \frac{Y}{X}$$





Ex.3 coordinates transformation (X, Y, Z) to (φ, λ, h)

Iterative procedure to estimate latitude and h

Step 1

$$r = \sqrt{X^2 + Y^2}$$

$$\varphi = \operatorname{arctg} \frac{Z}{r}$$

And an approx latitude:

Iterative procedure

Step 2

$$W = \sqrt{1 - e^2 \cdot \sin^2 \varphi}$$

$$N = \frac{a}{W}$$

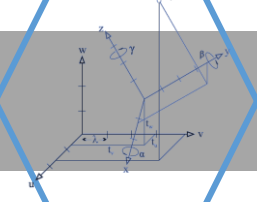
Step 3

$$h = \frac{X}{\cos \varphi \cos \lambda} - N$$

$$\varphi = \operatorname{arctg} \frac{Z}{r \left(1 - \frac{e^2 N}{N + h} \right)}$$

If $h(i) - h(i-1) < 10^{-8}$ AND $\varphi(i) - \varphi(i-1) < 10^{-8} \rightarrow$ **YES: STOP**

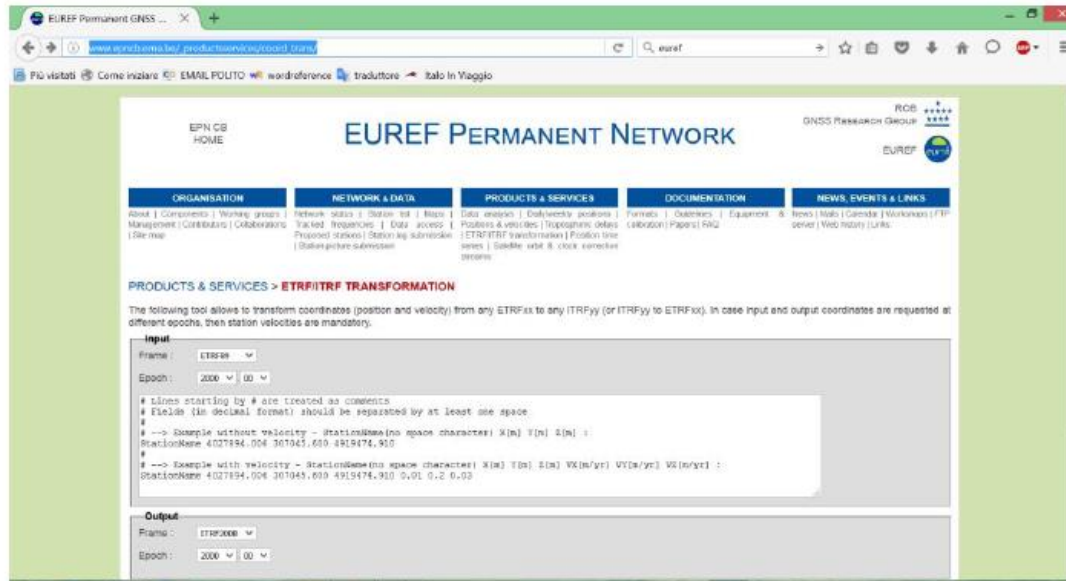
$\rightarrow$ **NO: repeat starting step 2**



Ex.4 reference system transformation

<http://www.epncb.oma.be/>

http://www.epncb.oma.be/_productsservices/coord_trans/



The screenshot shows the EUREF Permanent Network website. The main heading is "EUREF PERMANENT NETWORK". Below it, there are several tabs: ORGANISATION, NETWORK & DATA, PRODUCTS & SERVICES, DOCUMENTATION, and NEWS, EVENTS & LINKS. The "PRODUCTS & SERVICES" tab is selected, and the sub-tab "ETRF/ITRF TRANSFORMATION" is active. The page contains a form for transforming coordinates from ETRF to ITRF. The form has two main sections: "Input" and "Output". In the "Input" section, the "Frame" is set to "ETRF89" and the "Epoch" is set to "2000 00 00". Below this, there is a text area for inputting station data. It includes a comment line "# Lines starting by # are treated as comments" and a note "# Fields (in decimal format) should be separated by at least one space". There are two example lines: one for a station without velocity and one for a station with velocity. The "Output" section has a "Frame" dropdown set to "ITRF2000" and an "Epoch" dropdown set to "2000 00 00".

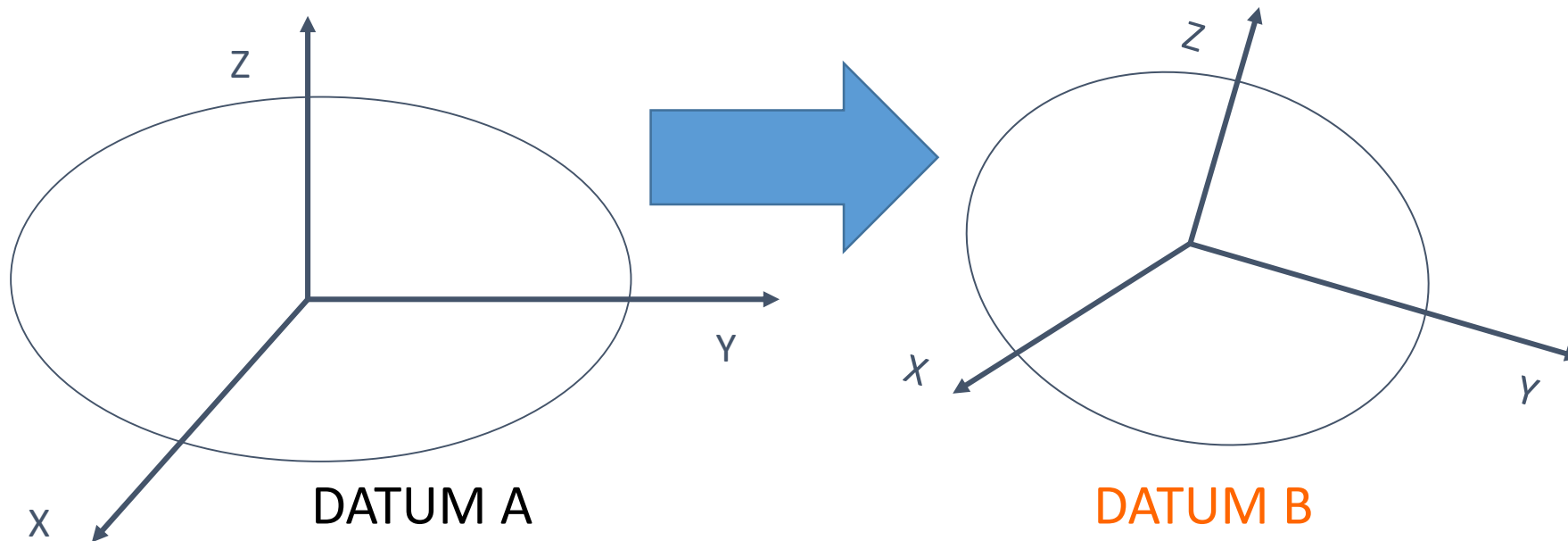
Using the XYZ of the EX. 2, to make the following transformations:

ETRF89 → ITRF89

ETRF2000 → ITRF2000

To analyze the differences.

Ex.5 Helmert's transformation



$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_B = \begin{pmatrix} X_0 \\ Y_0 \\ Z_0 \end{pmatrix} + \lambda \begin{bmatrix} 1 & R_z & -R_y \\ -R_z & 1 & R_x \\ R_y & -R_x & 1 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}_A$$

Ex.5 Helmert's transformation

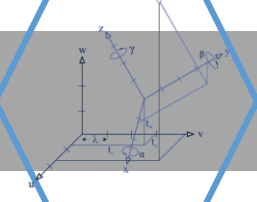
Linearizing the equation around the approx. values:

$$\begin{bmatrix} X_B - X_A - T_X^{(o)} \\ Y_B - Y_A - T_Y^{(o)} \\ Z_B - Z_A - T_Z^{(o)} \end{bmatrix} = \begin{bmatrix} \Delta T_X \\ \Delta T_Y \\ \Delta T_Z \end{bmatrix} + \begin{bmatrix} \Delta\lambda & R_Z & -R_X \\ -R_Z & \Delta\lambda & R_X \\ R_Y & -R_X & \Delta\lambda \end{bmatrix} \begin{bmatrix} X_A \\ Y_A \\ Z_A \end{bmatrix}$$

Solving the system with the least squares::

$$\begin{bmatrix} 1 & 0 & 0 & X_A & 0 & -Z_A & Y_A \\ 0 & 1 & 0 & Y_A & -Z_A & 0 & -X_A \\ 0 & 0 & 1 & Z_A & Y_A & X_A & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix} \cdot \begin{pmatrix} \Delta T_X \\ \Delta T_Y \\ \Delta T_Z \\ R_X \\ R_Y \\ R_Z \end{pmatrix} - \begin{bmatrix} X_B - X_A \\ Y_B - Y_A \\ Z_B - Z_A \\ \dots \end{bmatrix} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \dots \end{pmatrix}$$

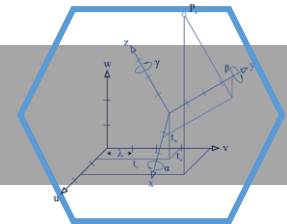
Ex.5 Helmert's transformation



$$\begin{matrix}
 \boxed{\begin{bmatrix} 1 & 0 & 0 & X_A & 0 & -Z_A & Y_A \\ 0 & 1 & 0 & Y_A & -Z_A & 0 & -X_A \\ 0 & 0 & 1 & Z_A & Y_A & X_A & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}} & \cdot & \boxed{\begin{pmatrix} \Delta T_X \\ \Delta T_Y \\ \Delta T_Z \\ \Delta\lambda \\ R_X \\ R_Y \\ R_Z \end{pmatrix}} & - & \boxed{\begin{bmatrix} X_B - X_A \\ Y_B - Y_A \\ Z_B - Z_A \\ \dots \end{bmatrix}} & = & \boxed{\begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ \dots \end{pmatrix}} \\
 \text{A} & & \text{X} & & \text{I}_0 & & \text{Residuals}
 \end{matrix}$$

$$X = (A^T A)^{-1} (A^T I_0) \quad \text{Least squares solution}$$

*NB: In order to avoid singular matrix, it is better to divide the coordinates (X, Y, Z) by 10⁶.
At the end, It is necessary to multiply the translations by this value.*



Ex.5 Helmert's transformation

To estimate the HELMERT PARAMETER using the points reported in «points_helmert.txt».

- 1) Parameters estimation*
- 2) Analysis of the residuals*

$$\text{Residuals } (v) = Ax - l_0$$

- 3) To make a comment of the results*